

SQL Constraints

Quick Revision & Exam Summary Notes

1. What are Constraints?

Constraints are conditions or restrictions applied to database columns (attributes). They control what data can be inserted into a table to ensure data accuracy and reliability.

2. UNIQUE

Does not allow duplicate values in a column.

Examples: Mobile Number, Email ID

💡 **Remember: No duplicates allowed**

3. NOT NULL

Does not allow empty/NULL values. The user must provide a value.

Important: NULL $\neq$ 0

- NULL: Value is absent/unknown
- 0: An actual numeric value

💡 **Remember: Must have a value**

4. PRIMARY KEY

★ High Yield

Uniquely identifies each row in a table.

Formula: UNIQUE + NOT NULL

Examples: Student (Roll Number), Employee (Employee ID)

💡 **Remember: UNIQUE + NOT NULL**

5. CHECK

Restricts values inserted into a column according to a specified logical condition.

```
CHECK (age > 18)
```

```
CHECK (address IN ('Delhi', 'Chandigarh', 'Mumbai'))
```

💡 **Remember: Condition / Domain restriction**

6. FOREIGN KEY

★ High Yield

Maintains **referential integrity** between tables by creating a relationship referencing another table's data.

💡 **Remember: Relationship + Referential Integrity**

7. DEFAULT

Provides a default value automatically when no value is explicitly supplied by the user.

```
salary INT DEFAULT 10000
```

💡 **Remember: Automatic default value**